PART I

For "penguins.csv" dataset.

1. Provide short details of the methods you used for data preprocessing.

A Methods that I used in data preprocessing are: Handling missing values and mismatched strings, detection and outlier handling, feature scaling of numerical columns, encoding of categorical features such as gender etc, feature correlation and selection.

2. a. Provide a brief overview of your dataset. Describe its domain, type of data, number of samples, and features.

Overview**:**

Domain: This dataset is used for research on penguin species, about their gender classification, and biological features.

Type of Data: It is a multivariate dataset with both categorial & numerical features.

Number of Samples:Dataset contains 344 penguins.

Features in the Dataset:

- species: Adelie, Chinstrap, Gentoo
- island: Dream, Torgersen, Biscoe
- bill_length_mm: length of the penguin's bill
- bill_depth_mm: depth of the penguin's bill
- flipper_length_mm: length of the penguin's flipper
- body_mass_g: penguin's body mass in grams
- gender: male, female

b. Present key statistics (e.g. mean, standard deviation, number of missing values for each feature)

Features	mean	Standard deviation	min	Max	Missing values
bill_length_mm	43.92	5.46	32.1	59.6	2
bill_depth_mm	17.15	1.97	13.1	21.5	2
flipper_length_mm	200.92	14.06	172	231	2
body_mass_g	4201.75	801.95	2700	6300	2
gender	-	-	-	-	11

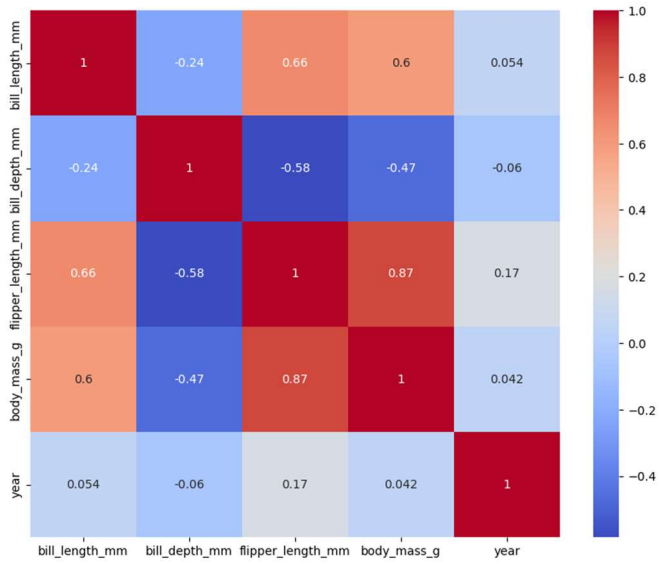
c. Include at least 5 graphs, such as histograms, scatter plots, or correlation matrices. Briefly describe the insights gained from these visualizations.

Data Visualizations and Insights

1. Correlation Heatmap:

The heatmap represents the correlation between the numerical features of penguin dataset. Dark red colors shows strong positive correlations whereas dark blue indicates strong negative correlations.

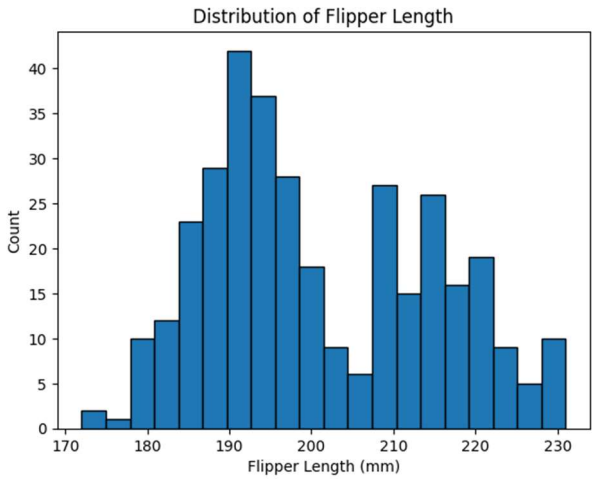
We can observe strong positive correlation between flipper_length_mm and body_mass_g. This indicates that larger penguins tend to have longer flippers. The correlation between bill_length_mm and flipper_length_mm is weaker.



2. Histogram of Flipper Length:

This histogram depicts the distribution of flipper lengths in the given dataset. The xaxis gives the flipper length , and the yaxis represent penguins count with those flipper lengths.

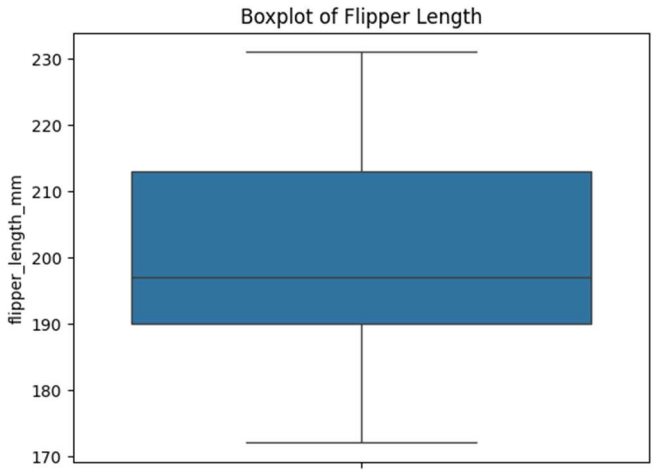
The distribution graph is normal with most penguins having flipper lengths between 180mm and 210mm.



3. Boxplot of Flipper Length:

The boxplot is used for detecting outliers in the flipper length feature. The box represents the interquartile range and the whiskers shows the range within 1.5 times IQR.

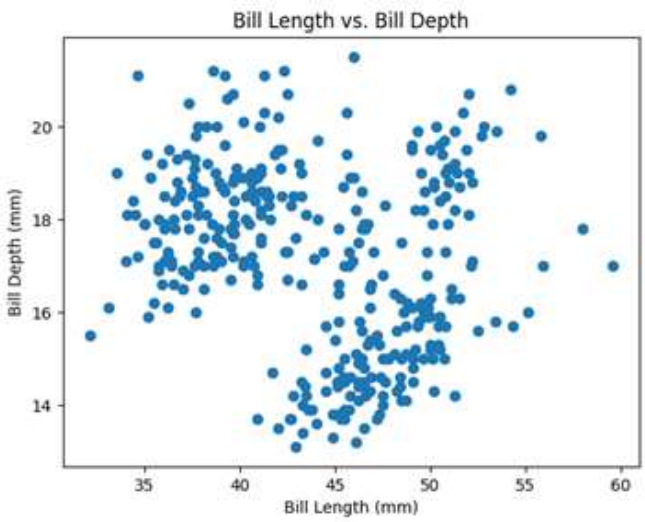
There are a few outliers in flipper length that indicates that only some penguins has particularly short or long flippers.



4. Scatter Plot of Bill Length vs Bill Depth:

This scatter plot gives the relationship between bill length and bill depth. Each point represents a penguin, with its bill length on the x-axis and bill depth on the y-axis.

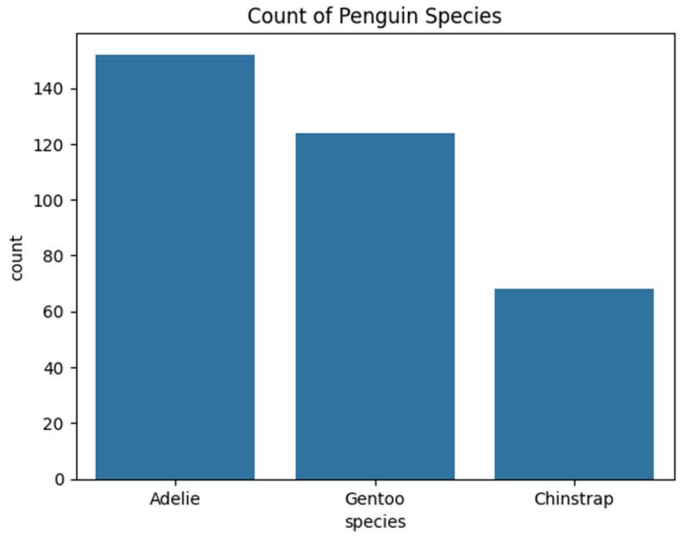
The scatter plot shows that there is no bill length and bill depth which indicates that these two features vary independently. This plot suggests that bill depth might provide different information than bill length when predicting 'species' or 'gender'.



5. Count Plot of Penguin Species:

The count plot visualizes the number of penguins per each species in the dataset. X-axis gives species,y-axis gives its count.

The graph shows that Adelie species are more in number where as chinstrap species have least species count.



<u>For "diamond.csv" dataset</u>.

1. Provide short details of the methods you used for data preprocessing.

I used following methods to clean and preprocess the dataset : Handling Missing Values,Data Type conversion, Feature encoding, Feature normalization.

2.

a) Provide a brief overview of your dataset. Describe its domain, type of data, number of samples, and features.

Overview:

Domain: Diamond Pricing Analysis

Type of Data: Mixed with Categorical and numerical data

Number of sample:

The dataset contains 53,930 samples after cleaning and oulier removal.

Features:

Catergorical:

- cut: Quality of the diamond cut (e.g., Fair, Good, Very Good, Premium, Ideal).
- color: Diamond color grade (ranging from D to J).
- clarity: Clarity of the diamond (e.g., I1, SI1, VS1, etc.).

Numerical:

- carat: Weight of the diamond.
- depth: Total depth percentage.
- table: Width of the diamond's table.
- price: Price of the diamond in USD.
- x: Length of the diamond in mm.
- y: Width of the diamond in mm.
- z: Height of the diamond in mm.

b) Present key statistics (e.g. mean, standard deviation, number of missing values for each feature)

Feature	Mean	Standard Deviation	Missing Values
Carat	0.7979	0.4740	0
Depth	61.75	1.432	0
Table	57.46	2.234	0
Price	3939.84	1544.19	0
x	5.7312	1.1223	0
y	5.7341	1.1218	0
z	3.5387	0.7057	0

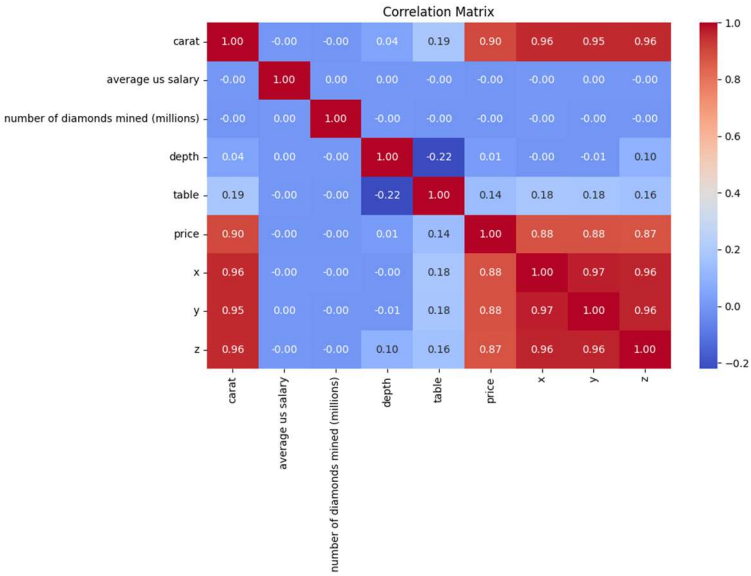
c. Include at least 5 graphs, such as histograms, scatter plots, or correlation matrices. Briefly describe the insights gained from these visualizations.

1. Carat and Price Correlation:

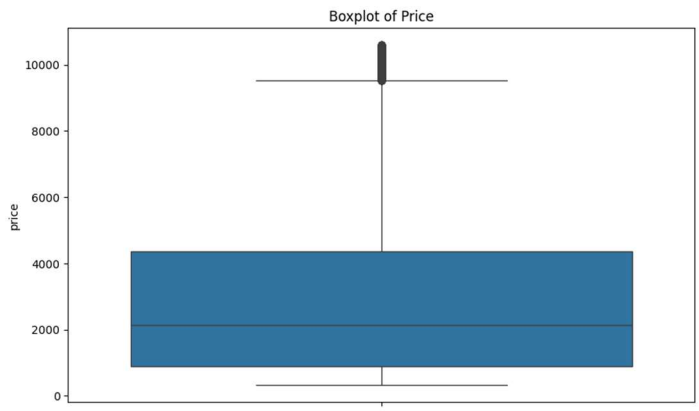
Heatmap shows the correlation matrix between different numerical data in our dataset.

From below heatmap we can observe that the carat and price have strong positive correlation (0.9)

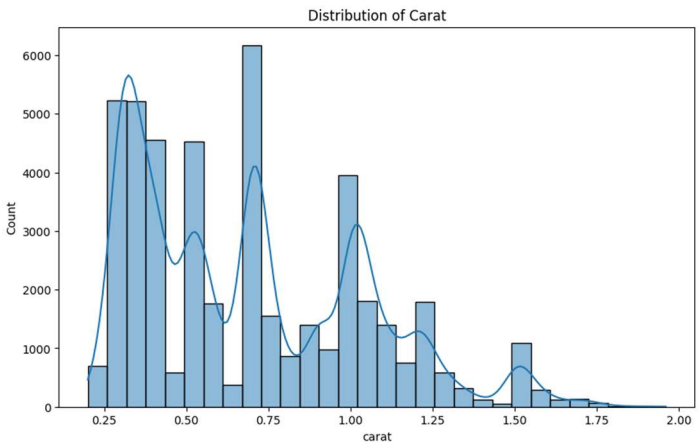
Carot and dimensions(x,y,z) are highly correlated (above 0.95).



2. Boxplot of Price**:** This plot is used for detecting outliers

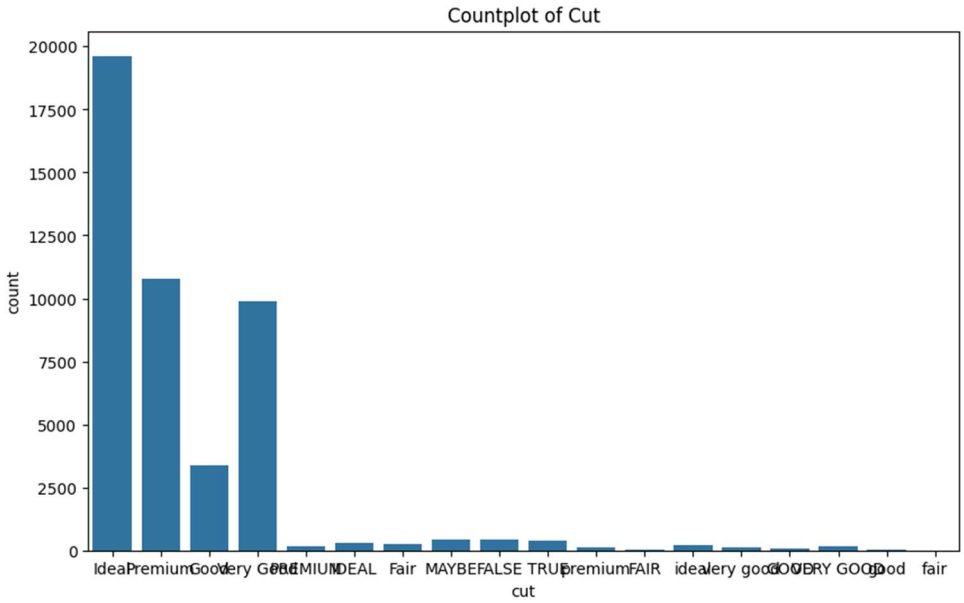


3. Count vs Carat Plot:
   This plot shows the distribution of the no. of diamonds for each carat value.



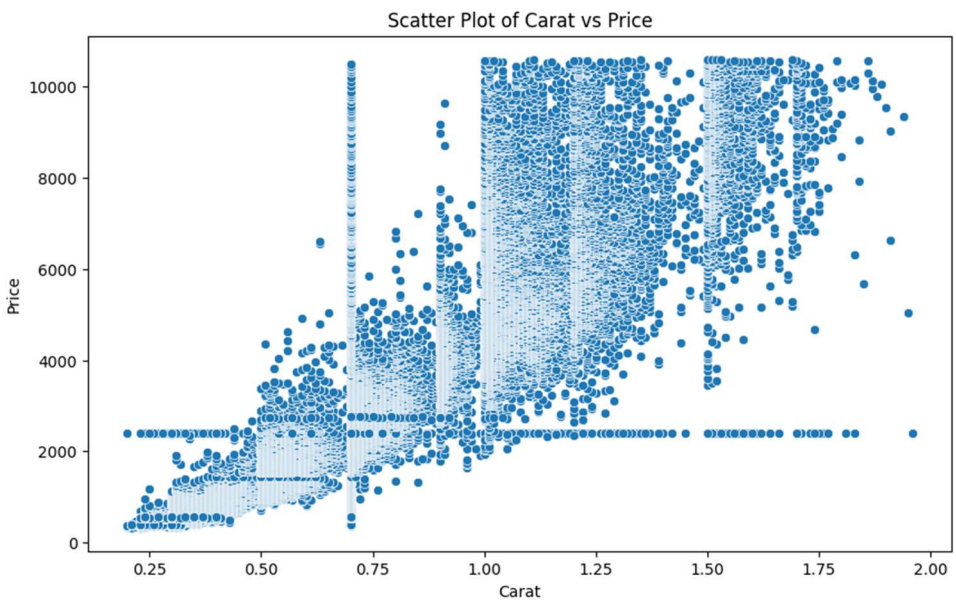
4. Countplot of ‘cut’:

   This plot shows the distribution of different cut qualities(i.e. good ,premium etc). It helps us to visualize how frequently each category of cuts appear in dataset. From graph we can observe that ‘Ideal cut’ is the most common cut.



5. Scatter plot of carat vs Price:

   This plot is used to visualise the relationship between diamond weight and their price.



<u>For "emissions_by_country.csv" dataset</u>.

1.Provide short details of the methods you used for data preprocessing.

I used following methods to clean and preprocess the dataset : Handling Missing Values,Data Type conversion, Feature encoding, Feature normalization.

2.

a) Provide a brief overview of your dataset. Describe its domain, type of data, number of samples, and features.

Overview:

Domain: Global Emission by different countries. It mainly focuses on various energy sources and their contribution to total emissions.

Type of Data: Mixed with Categorical and numerical data

Number of sample:

The dataset contains 63,104 samples.

Features:

- **Year**: The year in which the data was recorded (integer).
- **Total**: The total emissions for the given year (float).
- **Temperature**: Average temperature recorded for that year (float).
- **GDP Per Capita (USD)**: The gross domestic product per capita for the country (float).
- **Coal**: Emissions from coal (float).
- **Oil**: Emissions from oil (float).
- **Gas**: Emissions from natural gas (float).
- **Cement**: Emissions from cement production (float).
- **Flaring**: Emissions resulting from gas flaring (float).
- **Other**: Other sources of emissions not classified (float).
- **Per Capita**: Emissions per capita for the given country (float).

b) Present key statistics (e.g. mean, standard deviation, number of missing values for each feature)

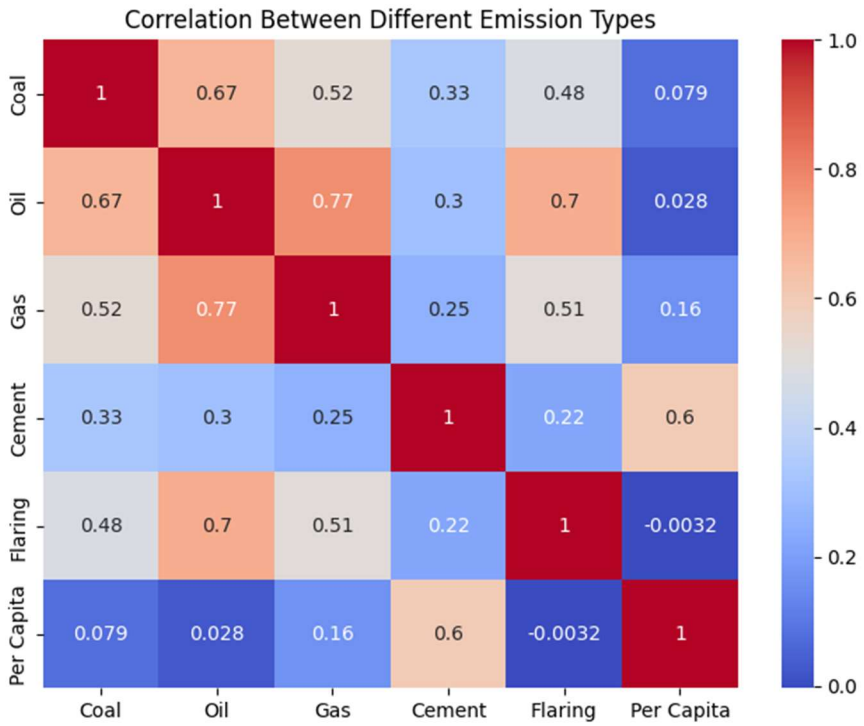
Table 1: Key Statistics of the Dataset

Feature	Count	Mean	Std Dev	Min	25%	50%	75%	Max
Year	62307	1888.27	122.65	1003	1816.00	1886.00	1955.00	2999
Total	62381	73.68	843.93	0.00	0.00	0.00	0.66	37123.85
Temperature	63104	49.50	17.29	20.00	35.00	49.00	64.00	79.00
GDP Per Capita (USD)	63104	39026.54	10975.54	20000.00	29498.75	39067.00	48544.25	57999.00
Coal	21797	127.39	677.95	0.00	0.00	0.34	8.50	15051.51
Oil	21774	153.48	670.83	0.00	0.11	1.34	13.01	12345.65
Gas	21717	125.16	514.39	0.00	0.00	0.00	1.68	7921.83
Cement	20488	62.60	353.92	0.00	0.00	0.03	0.73	2982.00
Flaring	21338	56.07	337.63	0.00	0.00	0.00	0.00	2994.00
Other	2685	849.40	1055.25	0.00	1.02	9.24	1812.00	2999.00
Per Capita	19392	121.57	489.34	0.00	0.23	1.60	6.31	2997.00

c. Include at least 5 graphs, such as histograms, scatter plots, or correlation matrices. Briefly describe the insights gained from these visualizations.

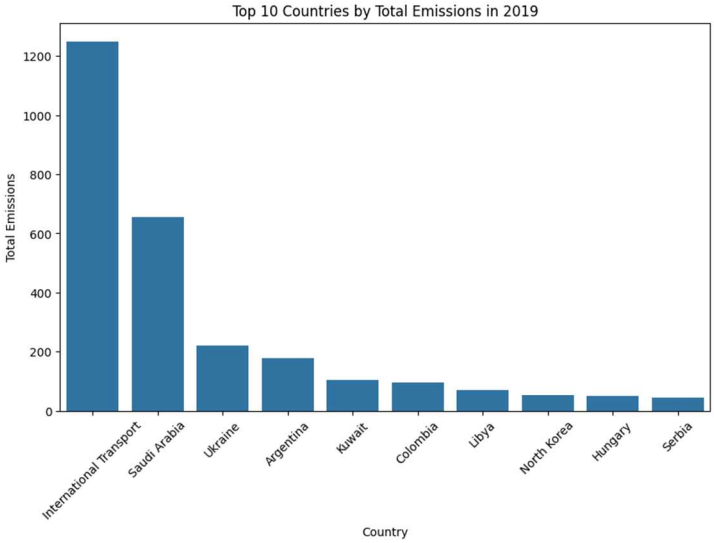
1. Heatmap :

Heatmap of the correlation matrix between different emission sources shows how closely these sources are related. High correlation values means that some emission types tend to rise and fall together, indicating their possible interdependence. Emissions from Coal and Oil may show a strong positive correlation.



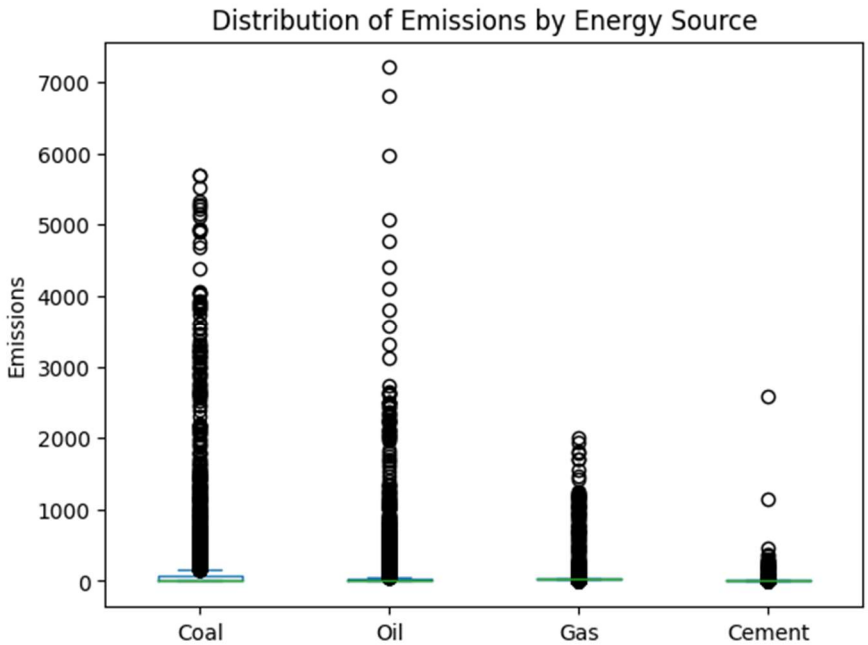
2. Bar Plot of Top 10 Countries by Total Emissions in 2019

This Bar graph highlights the total emissions in the 2019 by the top 10 countries. It clearly shows which nations are largest contributors to global emissions. From the graph we can observe that International transport and Saudi Arabia are the largest contributes for global emission.



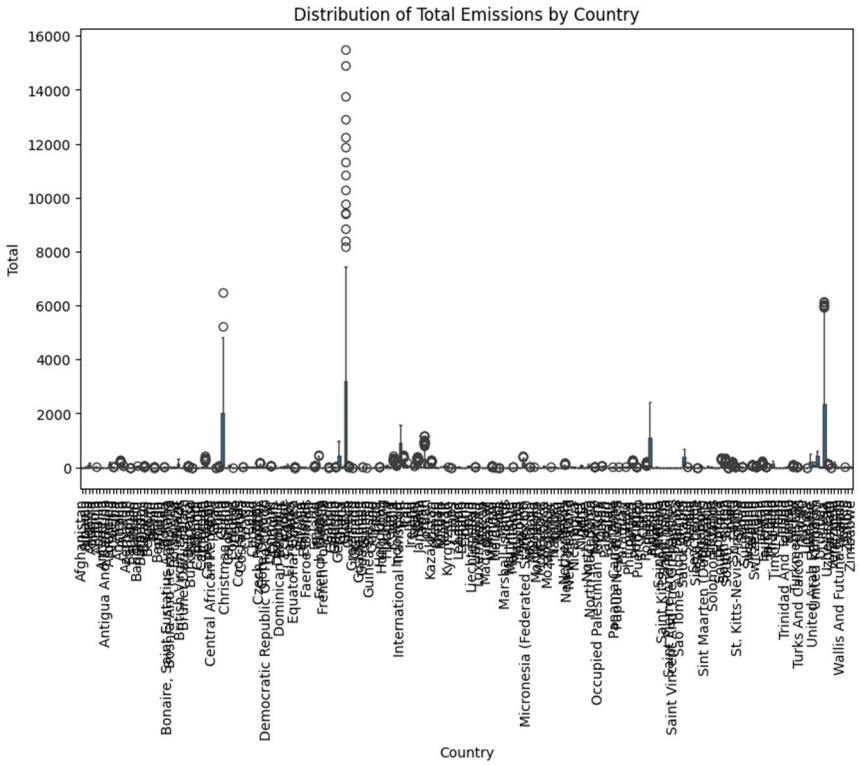
3. Boxplot for Distribution of Emissions by Energy Source:

Emissions from different energy sources is illustrated in the Boxplot. Coal have the widest range of emissions according to the plot.



4. Boxplot for Total Emissions by Country

This graph shows the distribution of emissions per country. This visualization highlights the unequal distribution of global emissions.



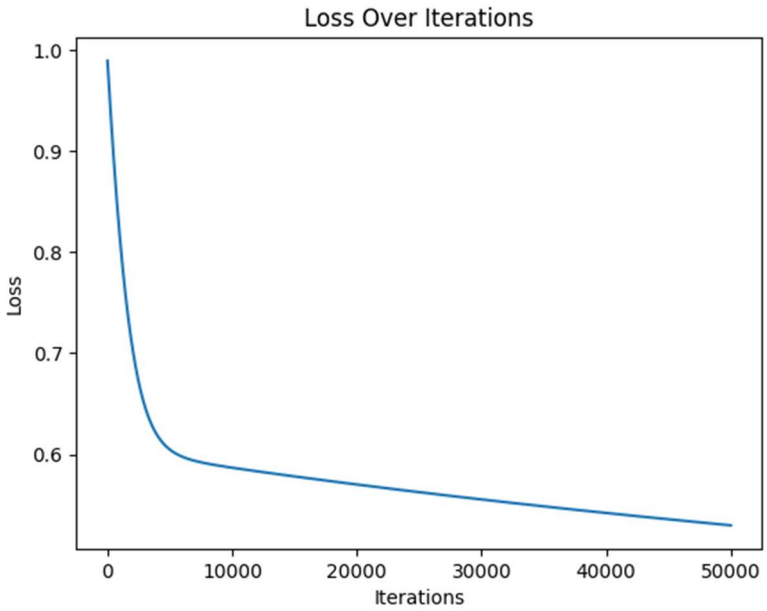
PART II

For “penguins.csv” dataset.

1. Provide your best accuracy.

A. Best accuracy for the penguin dataset with the implemented model is 89.85

2. Include a loss graph and provide a short analysis of the results. CSE 574-D: Introduction to Machine Learning, Fall 2024



From the above graph we can say that the gradient descent is working correctly because the loss function is decreasing steadily with the increase in number of iterations.

3. Provide at least 3 different setups with learning rate and #iterations and discuss the results along with plotting of graphs. Plot the graph to discuss impact and loss over the iterations. Explain how hyperparameters influence the accuracy of the model.
I have considered learning rates 0.001,0.01,0.0001 for 50000,100000,150000 iterations respectively. I observed that with learning rate 0.001 the model achieved best accuracy results compared to remaining learning rates.

4. Discuss the benefits/drawbacks of using a Logistic Regression model.

Benefits:

- Logistic regression is easy to implement and it also supports regularization.
- Its efficiency is more for binary classification tasks.

Drawbacks:

- It is sensitive to outliers and Multicollinearity.

PART-III

1. Provide a brief description of the dataset you used, including its characteristics, number of samples, and features.

Dataset Description:

This dataset is mainly used for analysing global emissions and various environmental factors,

The characteristics of the dataset:

Domain: Global Emission by different countries. It mainly focuses on various energy sources and their contribution to total emissions.

Type of Data: Mixed with Categorical and numerical data

Number of sample:

The dataset contains 63,104 samples.

Features:

- Year: The year in which the data was recorded (integer).
- Total: The total emissions for the given year (float).
- Temperature: Average temperature recorded for that year (float).
- GDP Per Capita (USD): The gross domestic product per capita for the country (float).
- Coal: Emissions from coal (float).
- Oil: Emissions from oil (float).
- Gas: Emissions from natural gas (float).
- Cement: Emissions from cement production (float).
- Flaring: Emissions resulting from gas flaring (float).
- Other: Other sources of emissions not classified (float).
- Per Capita: Emissions per capita for the given country (float).

2. Model Evaluation:

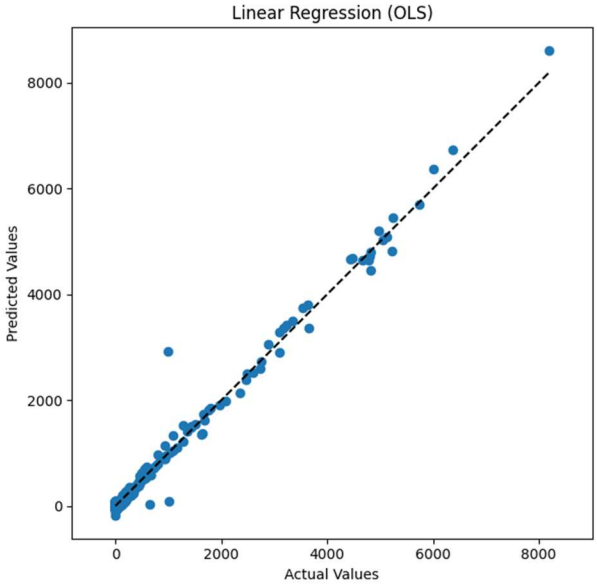
Loss Value: Performance of both the linear regression and ridge regression model was evaluated using the Mean Squared Error (MSE):

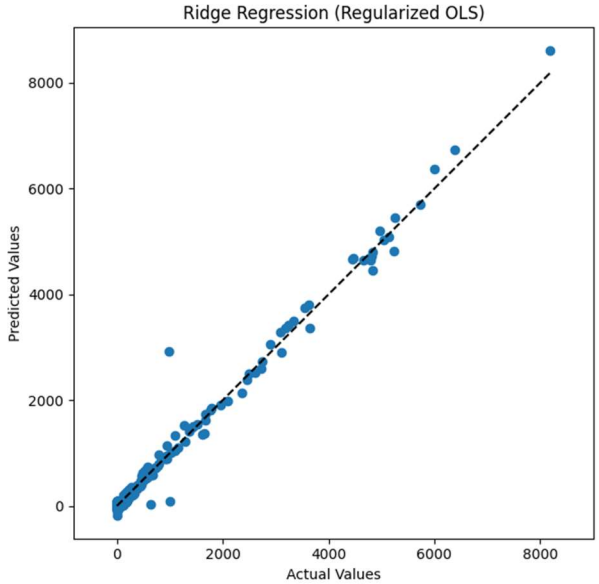
MSE for Linear Regression (OLS) & Ridge Regression(Ridge):

```
MSE (OLS): 3383.049253523946
MSE (Ridge): 3383.0492524227575
```

Predictions vs Actual

The following scatter plots compare predicted values against actual values for both linear regression and ridge regression:





In these plots the X-axis represents actual values of total emissions, while the Y-axis shows predicted values. The diagonal line indicates perfect predictions which helps in visualizing the accuracy of the models.

3. Discussion

Benefits and Limitations of OLS

Benefits:

- OLS provides an intuitive and direct method for estimating linear relationships between independent variables and the target variable.
- This method would yield optimal weights estimates under the fulfillment of assumptions, such as homoscedasticity and no multicollinearity.

Limitations:

- OLS approach is sensitive to outliers which can significantly skew the results.
- It may also lead to overfitting if the model complexity is too high relative to the amount of training data.

Strengths and Weaknesses of Linear Regression

Strengths:

- Linear regression is easy to interpret and implement.
- The computational efficiency of linear regression is more even for large datasets.

Weaknesses:

- It assumes a linear relationship which may not hold true in all cases.
- It is vulnarable to overfitting in the presence of high-dimensional data.

Motivation for L2 Regularization and Ridge Regression

Main motivation for Ridge regression is that it solves the drawbacks of linear regression.Ridge regression employs l2 regularization which adds a penalty that is equal to square root of the magnitude of coeffs to loss function,

Benefits:

- It reduces overfitting by constraining the weights and leads to a more generalizable model.
- It effectively handles multicollinearity effectively by distributing the coefficient values more evenly across correlated predictors.

Drawbacks:

- This regression does not perform variable selection. All features are retained in the model itself.
- It isa challenging task for choosing the optimal value for the regularization parameter lambda and may require cross-validation.

PART-IV

1.Provide a brief description of the dataset you used, including its characteristics, number of samples, and features.

Overview:The dataset is used for analysis of the diamonds dataset, which consists of information related to various characteristics of diamonds and their prices.

Domain: Diamond Pricing Analysis

Type of Data: Mixed with Categorical and numerical data

Number of sample:

The dataset contains 53,930 samples after cleaning and oulier removal.

Features:

Catergorical:

- cut: Quality of the diamond cut (e.g., Fair, Good, Very Good, Premium, Ideal).
- color: Diamond color grade (ranging from D to J).
- clarity: Clarity of the diamond (e.g., I1, SI1, VS1, etc.).

Numerical:

- carat: Weight of the diamond.
- depth: Total depth percentage.
- table: Width of the diamond's table.
- price: Price of the diamond in USD.
- x: Length of the diamond in mm.
- y: Width of the diamond in mm.
- z: Height of the diamond in mm.

2, Data Analysis and Preprocessing:

- Handling Missing Values: Missing values are replace by using mean ,median and mode based on the requirements.
- Target Variable: The target variable is isolated and a feature matrix was created .
- Dropped categorical Variable: Categorical features such as cut, color, clarity were dropped.

Train-Test Split:

The dataset was split into training(80%) and testing(20%) sets resulting in following shapes:

Shape of X_train: (43152, 8)

Shape of Y_train: (43152, 1)

Shape of X_test: (10788, 8)

Shape of Y_test: (10788, 1)

Feature Normalization is implemented to enhance the predictive performance.

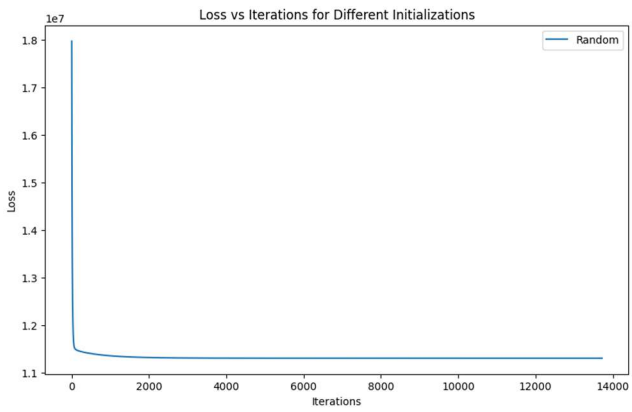
2.Model Evaluation:

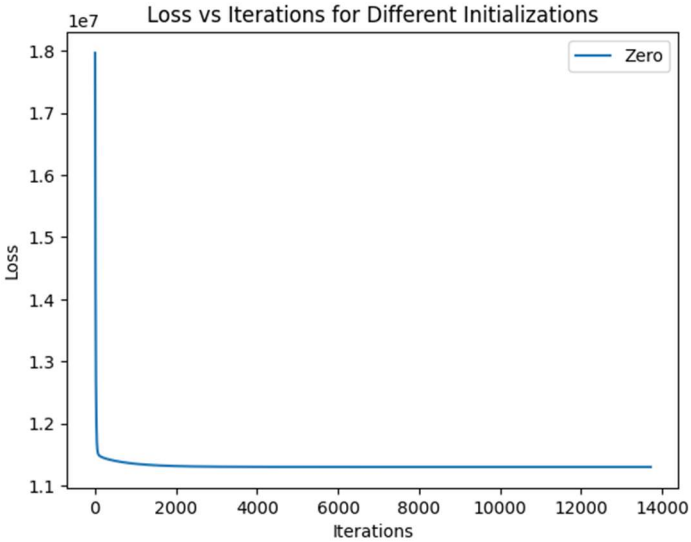
Loss value: Loss results for the model are

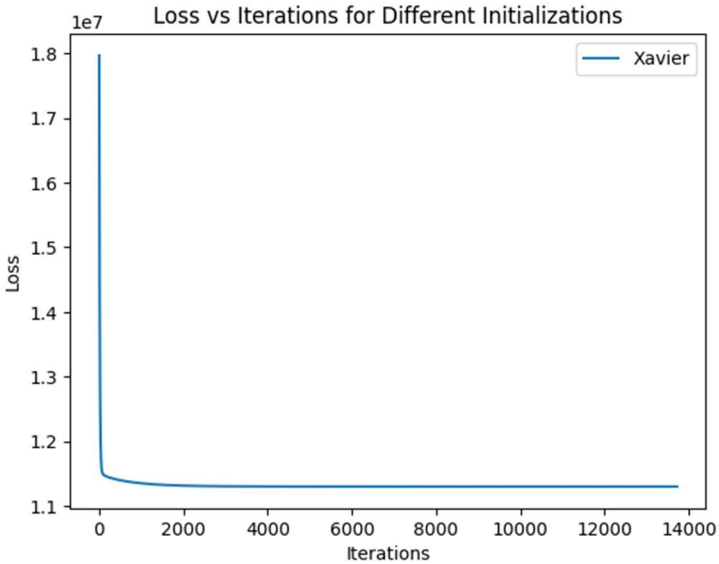
```
Training with Random initialization:
Iteration 0, Loss: 17966937.55250581
Iteration 1000, Loss: 11597181.950515848
Iteration 2000, Loss: 11594551.50670814
Iteration 3000, Loss: 11594453.664557125
Iteration 4000, Loss: 11594449.86798642
Iteration 5000, Loss: 11594449.712777171
Converge at iteration 5832
Final training loss: 11594449.70626331
Test loss: 6887036.652451084

Training with Zero initialization:
Iteration 0, Loss: 17966928.83795028
Iteration 1000, Loss: 11597181.952232076
Iteration 2000, Loss: 11594551.506611412
Iteration 3000, Loss: 11594453.664545784
Iteration 4000, Loss: 11594449.867985614
Iteration 5000, Loss: 11594449.712777123
Converge at iteration 5832
Final training loss: 11594449.706263306
Test loss: 6887036.652478649

Training with Xavier initialization:
Iteration 0, Loss: 17963857.684142143
Iteration 1000, Loss: 11597183.082748929
Iteration 2000, Loss: 11594551.5853824
Iteration 3000, Loss: 11594453.669423375
Iteration 4000, Loss: 11594449.86827285
Iteration 5000, Loss: 11594449.712793684
Converge at iteration 5832
Final training loss: 11594449.706264842
Test loss: 6887036.633852796
```







Weight Initialization methods:

3 different weight initialization methods were used:

- **Random Initialization**: Weights were initialized randomly which allows diverse starting points but potentially cause varied convergence rates.
- **Zero Initialization**: All weights were initialized to zero which leads to poor performance as it may cause symmetry that prevents the model from learning.
- **Xavier Initialization**: Weights were initialized based on the number of input and output neurons which provide a balanced starting point that maintains variance in activation.

Stopping Criteria:

Fixed iterations and Gradient Threshold were used as stopping criteria.

3. Discussion on Elastic Net Regularization:

Elastic Net Regularization combines the advantages of L1 and l2 regularization.L1 promotes feature selection and L2 prevents overfitting.

It can select multiple features that are correlated with each other.

Benefits:

- Improved Prediction Accuracy
- Enhanced model interpretability because of feature selection

Drawbacks:

- Tuning of regularization parameters lambda 1 and lambda 2 is a challenging task
- It requires longer computation time compared to simple regression models.